

EXTENSION 1

Stage 6

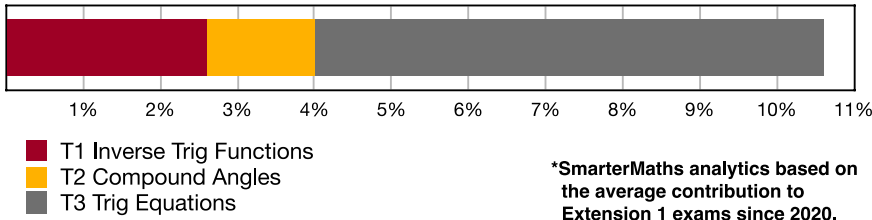
Trigonometry (Ext1)

T1 Inverse Trig Functions (Y11)

Teacher: SHS Maths

Exam Equivalent Time: 91.5 minutes (based on allocation of 1.5 minutes per mark)

T1-T3 Trigonometry



HISTORICAL CONTRIBUTION

- Trigonometry* has contributed a very healthy 10.6%, on average, to new syllabus Ext1 exams since being introduced in 2020.
- Trigonometry is split into three topic areas for analysis purposes which are: *T1-Inverse Trig Functions* (2.6%), *T2-Compound Angles* (1.4%) and *3-Trig Equations* (6.6%).
- This analysis looks at *T1 Inverse Trig Functions*.

HSC ANALYSIS - What to expect and common pitfalls

- Inverse Trig Functions* (2.6%) has been examined in two dedicated questions each year between 2022-24.
- "Dedicated question" is used above as this topic area also commonly appears in difficult cross-topic calculus questions that are tested towards the end of the Ext1 exam (not included here).
- Drawing or identifying an inverse trig graph is a common question type, last examined in 2024.
- A deep understanding of domain and range of an inverse function is critical and regularly tested (see *2022 Ext1 13c* and *2024 Ext1 4 MC*).
- Exact values of inverse functions was tested in *2023 Ext1 5 MC* and *2022 Ext1 1 MC*, proving challenging on both occasions.
- Note the syllabus explicitly states that students must understand the notation *arcsin* etc., which was required in 2023 and 2022 and features in numerous database questions.

Questions

1. Trigonometry, EXT1 T1 2015 HSC 6 MC

What is the domain of the function $f(x) = \sin^{-1} (2x)$?

- A. $-\pi \leq x \leq \pi$
- B. $-2 \leq x \leq 2$
- C. $-\frac{\pi}{4} \leq x \leq \frac{\pi}{4}$
- D. $-\frac{1}{2} \leq x \leq \frac{1}{2}$

2. Trigonometry, EXT1 T1 2024 HSC 5 MC

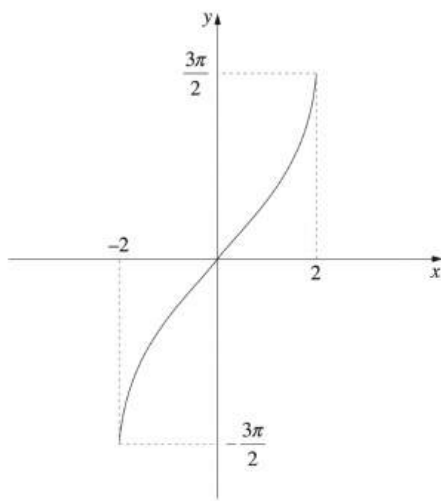
Consider the function $g(x) = 2\sin^{-1} (3x)$.

Which transformations have been applied to $f(x) = \sin^{-1} (x)$ to obtain $g(x)$?

- A. Vertical dilation by a factor of $\frac{1}{2}$ and a horizontal dilation by a factor of $\frac{1}{3}$
- B. Vertical dilation by a factor of $\frac{1}{2}$ and a horizontal dilation by a factor of 3
- C. Vertical dilation by a factor of 2 and a horizontal dilation by a factor of $\frac{1}{3}$
- D. Vertical dilation by a factor of 2 and a horizontal dilation by a factor of 3

3. Trigonometry, EXT1 T1 2012 HSC 4 MC

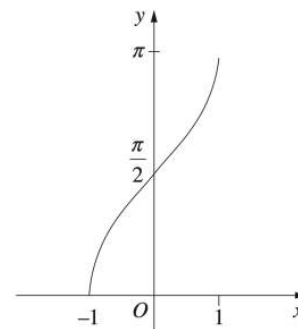
Which function best describes the following graph?



- A. $y = 3\sin^{-1}2x$
- B. $y = \frac{3}{2}\sin^{-1}2x$
- C. $y = 3\sin^{-1} \frac{x}{2}$
- D. $y = \frac{3}{2}\sin^{-1} \frac{x}{2}$

4. Trigonometry, EXT1 T1 2013 HSC 9 MC

The diagram shows the graph of a function.



Which function does the graph represent?

- A. $y = \cos^{-1}x$
- B. $y = \frac{\pi}{2} + \sin^{-1}x$
- C. $y = -\cos^{-1}x$
- D. $y = -\frac{\pi}{2} - \sin^{-1}x$

5. Trigonometry, EXT1 T1 2017 HSC 7 MC

Which diagram represents the domain of the function $f(x) = \sin^{-1}\left(\frac{3}{x}\right)$?

- A.
- B.
- C.
- D.

6. Trigonometry, EXT1 T1 2022 HSC 1 MC

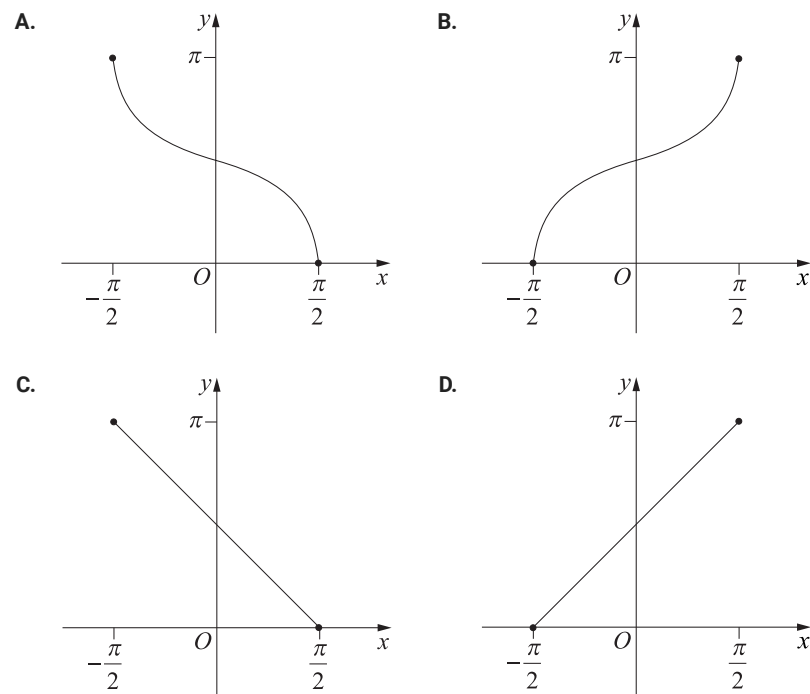
It is given that $\cos\left(\frac{23\pi}{12}\right) = \frac{\sqrt{6} + \sqrt{2}}{4}$.

Which of the following is the value of $\cos^{-1}\left(\frac{\sqrt{6} + \sqrt{2}}{4}\right)$?

- A. $\frac{23\pi}{12}$
- B. $\frac{11\pi}{12}$
- C. $\frac{\pi}{12}$
- D. $-\frac{11\pi}{12}$

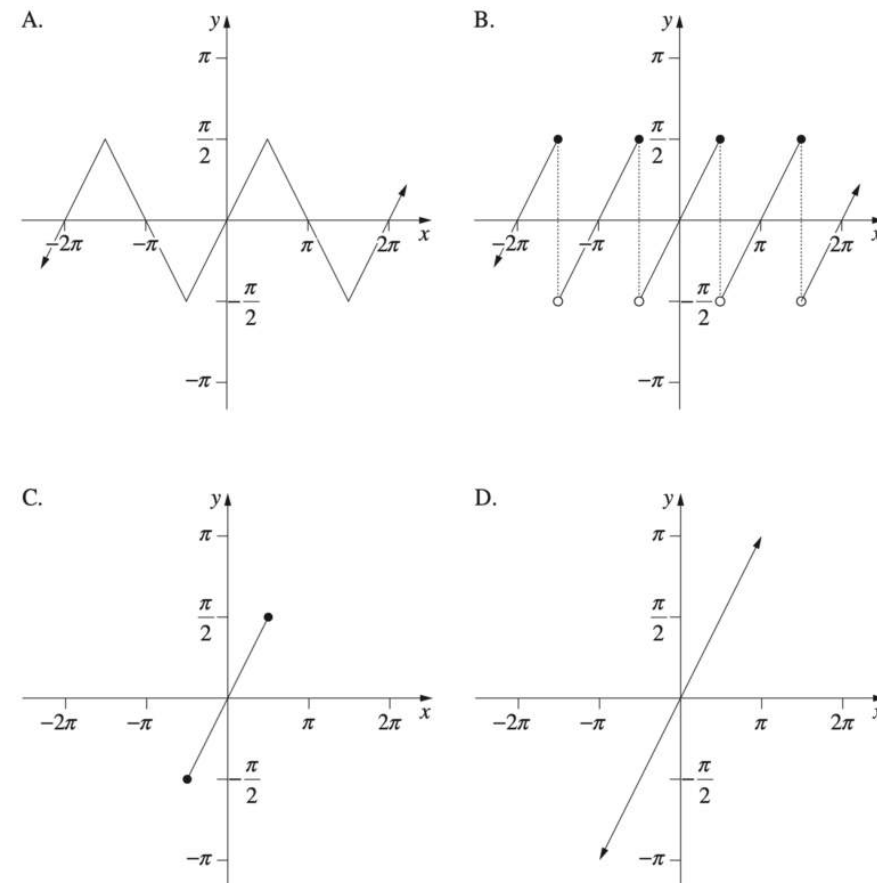
7. Trigonometry, EXT1 T1 2019 HSC 9 MC

Which graph best represents $y = \cos^{-1}(-\sin x)$, for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$?



8. Trigonometry, EXT1 T1 2021 HSC 9 MC

Which graph represents the function $y = \sin^{-1}(\sin x)$?



9. Trigonometry, EXT1 T1 2023 HSC 5 MC

Which of the following is the value of $\sin^{-1}(\sin a)$ given that $\pi < a < \frac{3\pi}{2}$?

- A. $a - \pi$
- B. $\pi - a$
- C. a
- D. $-a$

10. Trigonometry, EXT1 T1 2023 HSC 7 MC

Which statement is always true for real numbers a and b where $-1 \leq a < b \leq 1$?

- A. $\sec a < \sec b$
- B. $\sin^{-1} a < \sin^{-1} b$
- C. $\arccos a < \arccos b$
- D. $\cos^{-1} a + \sin^{-1} a < \cos^{-1} b + \sin^{-1} b$

11. Trigonometry, EXT1 T1 2024 HSC 4 MC

What are the domain and range of the function $y = 2\cos^{-1}(2x) + 2\sin^{-1}(2x)$?

- A. Domain: $[-0.5, 0.5]$ and Range: $\{\pi\}$
- B. Domain: $[-0.5, 0.5]$ and Range: $[-\pi, 3\pi]$
- C. Domain: $[-2, 2]$ and Range: $\{\pi\}$
- D. Domain: $[-2, 2]$ and Range: $[-\pi, 3\pi]$

12. Trigonometry, EXT1 T1 EQ-Bank 3 MC

The graph of the function $y = \arccos(x - 3)$ is transformed by being dilated horizontally with a scale factor of $\frac{1}{2}$ and then translated to the left by 1.

What is the equation of the transformed graph?

- A. $y = \cos^{-1}\left(\frac{x-5}{2}\right)$
- B. $y = \cos^{-1}\left(\frac{x-8}{2}\right)$
- C. $y = \cos^{-1}(2x-1)$
- D. $y = \cos^{-1}(2x-2)$

13. Trigonometry, EXT1 T1 EQ-Bank 4 MC

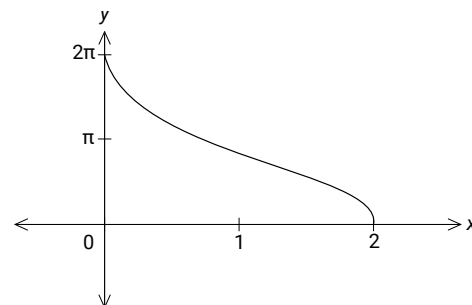
The graph of the function $y = \sin^{-1}(x - 2)$ is transformed by being dilated by a factor of 3 from the y -axis and then translated to the right by 2.

What is the equation of the transformed graph?

- A. $y = \sin^{-1}\left(\frac{x-8}{3}\right)$
- B. $y = \sin^{-1}\left(\frac{x-12}{3}\right)$
- C. $y = \sin^{-1}(3x-4)$
- D. $y = \sin^{-1}(3x-8)$

14. Trigonometry, EXT1 T1 EQ-Bank 5 MC

Which equation describes the graph shown?



- A. $y = 2\cos^{-1}(x+1)$
- B. $y = \sin^{-1}(x-1)$
- C. $y = 2\cos^{-1}(x-1)$
- D. $y = \frac{1}{2}\sin^{-1}(x+1)$

15. Trigonometry, EXT1 T1 EQ-Bank 2

State the domain and range of the function

$$y = \arccos 3x \quad (2 \text{ marks})$$

16. Trigonometry, EXT1 T1 2011 HSC 1e

Find the exact value of $\cos^{-1}\left(-\frac{1}{2}\right)$. (1 mark)

17. Trigonometry, EXT1 T1 2005 HSC 1c

State the domain and range of $y = \cos^{-1}\left(\frac{x}{4}\right)$. (2 marks)

18. Trigonometry, EXT1 T1 2017 HSC 11d

Sketch the graph of the function $y = 2\cos^{-1}x$. (2 marks)

19. Trigonometry, EXT1 T1 2014 HSC 11c

Sketch the graph $y = 6\tan^{-1}x$, clearly indicating the range. (2 marks)

20. Trigonometry, EXT1 T1 EQ-Bank 1

Show that $\cos(\sin^{-1}q) = \sqrt{1-q^2}$ (2 marks)

21. Trigonometry, EXT1 T1 SM-Bank 2

Evaluate β if $\beta = \cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)$. (2 marks)

22. Calculus, EXT1 C2 2006 HSC 2a

Let $f(x) = \sin^{-1}(x + 5)$.

- i. State the domain and range of the function $f(x)$. (2 marks)
- ii. Find the gradient of the graph of $y = f(x)$ at the point where $x = -5$. (2 marks)
- iii. Sketch the graph of $y = f(x)$. (2 marks)

23. Trigonometry, EXT1 T1 2007 HSC 2b

Let $f(x) = 2\cos^{-1}x$.

- i. Sketch the graph of $y = f(x)$, indicating clearly the coordinates of the endpoints of the graph. (2 marks)
- ii. State the range of $f(x)$. (1 mark)

24. Trigonometry, EXT1 T1 EQ-Bank 5

- a. State the domain and range for $f(x) = 4\sin^{-1}(2x - 3)$. (2 marks)
- b. Sketch the function $f(x) = 4\sin^{-1}(2x - 3)$. (1 mark)

25. Trigonometry, EXT1 T1 2007 HSC 5c

Find the exact values of x and y which satisfy the simultaneous equations

$$\sin^{-1} x + \frac{1}{2} \cos^{-1} y = \frac{\pi}{3} \text{ and}$$
$$3 \sin^{-1} x - \frac{1}{2} \cos^{-1} y = \frac{2\pi}{3}. \text{ (3 marks)}$$

26. Trigonometry, EXT1 T1 2010 HSC 1b

Let $f(x) = \cos^{-1}\left(\frac{x}{2}\right)$. What is the domain of $f(x)$? (1 mark)

27. Trigonometry, EXT1 T1 2011 HSC 2d

Sketch the graph of the function $f(x) = 2\arccos x$. Clearly indicate the domain and range of the function. (2 marks)

28. Trigonometry, EXT1 T1 2012 HSC 13a

Write $\sin\left(2\cos^{-1}\left(\frac{2}{3}\right)\right)$ in the form $a\sqrt{b}$, where a and b are rational. (2 mark)

29. Trigonometry, EXT1 T1 EQ-Bank 4

Sketch the graph $y = 2\cos^{-1}(x + 1)$. (3 marks)

30. Trigonometry, EXT1 T1 SM-Bank 1

Find $\sin\theta$ given that $\theta = \arccos\left(\frac{12}{13}\right) + \arctan\left(\frac{3}{4}\right)$, and $0 \leq \theta \leq \frac{\pi}{2}$. (3 marks)

31. Trigonometry, EXT1 T1 SM-Bank 3

Determine the exact value of

$$\sin^{-1}\left(\frac{\sqrt{3}}{2}\right) - \sin^{-1}\left(-\frac{\sqrt{3}}{2}\right) - \cos^{-1}\left(-\frac{1}{2}\right). \text{ (3 marks)}$$

32. Trigonometry, EXT1 T1 SM-Bank 4

Sketch $y = 3\cos^{-1}(2x - 1)$ (3 marks)

33. Trigonometry, EXT1 T1 2022 HSC 13c

The function f is defined by $f(x) = \sin(x)$ for all real numbers x . Let g be the function defined on $[-1, 1]$ by $g(x) = \arcsin(x)$.

Is g the inverse of f ? Justify your answer. (2 marks)

Worked Solutions

1. Trigonometry, EXT1 T1 2015 HSC 6 MC

f(x) = sin^-1 (2x)

Domain of f(x) = sin^-1 x is

-1 ≤ x ≤ 1

∴ Domain of f(x) = sin^-1 (2x) is

-1 ≤ 2x ≤ 1

-1/2 ≤ x ≤ 1/2

⇒ D

2. Trigonometry, EXT1 T1 2024 HSC 5 MC

A vertical dilation of factor 2:

f(x) = sin^-1(x) → f_1(x) = 2sin^-1(x)

A horizontal dilation of factor 1/3:

f_1(x) = 2sin^-1(x) → f_2(x) = 2sin^-1(3x)

⇒ C

3. Trigonometry, EXT1 T1 2012 HSC 4 MC

Domain: -2 ≤ x ≤ 2

-1 ≤ x/2 ≤ 1

∴ Graph is y = asin^-1 x/2

When x = 2, y = 3π/2:

3π/2 = asin^-1 1

3π/2 = a × π/2

a = 3

∴ y = 3sin^-1 x/2

⇒ C

Worked Solutions

4. Trigonometry, EXT1 T1 2013 HSC 9 MC

By elimination,

The graph passes through (1, π)

The only equation to satisfy this point is

y = π/2 + sin^-1 x

⇒ B

5. Trigonometry, EXT1 T1 2017 HSC 7 MC

f(x) = sin^-1(3/x)

3/x ≥ -1 and 3/x ≤ 1

x/3 ≤ -1 x ≥ 3

x ≤ -3

⇒ A

6. Trigonometry, EXT1 T1 2022 HSC 1 MC

23π/12 ⇒ 4th quadrant

cos(23π/12) = cos(2π - 23π/12) = cos(π/12)

∴ cos^-1((√6 + √2)/4) = π/12

⇒ C

Mean mark 54%.

7. Trigonometry, EXT1 T1 2019 HSC 9 MC

y = cos^-1(-sin x)

= cos^-1(cos(x + π/2))

= x + π/2

⇒ D

8. Trigonometry, EXT1 T1 2021 HSC 9 MC

By elimination:

$$\text{At } x = \pi, y = \sin^{-1}(\sin \pi) = \sin^{-1} 0 = 0$$

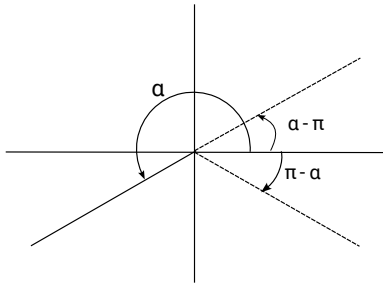
→ Eliminate C and D

$$\begin{aligned} \text{At } x = \frac{2\pi}{3}, y &= \sin^{-1}\left(\sin \frac{2\pi}{3}\right) \\ &= \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) \\ &= \frac{\pi}{3} \end{aligned}$$

→ Eliminate B

⇒ A

9. Trigonometry, EXT1 T1 2023 HSC 5 MC



$$\sin(a) = -\sin(a - \pi) \text{ (see above)}$$

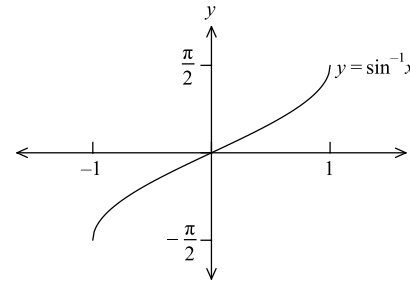
Mean mark 56%.

$$\begin{aligned} \sin^{-1}(\sin(a)) &= \sin^{-1}(-\sin(a - \pi)) \\ &= \sin^{-1}(\sin(\pi - a)) \\ &= \pi - a \end{aligned}$$

⇒ B

10. Trigonometry, EXT1 T1 2023 HSC 7 MC

Consider the graph of $y = \sin^{-1} x$



$y = \sin^{-1} x$ is an increasing function in range $-1 \leq x \leq 1$

∴ Given $-1 \leq a < b \leq 1 \Rightarrow \sin^{-1} a < \sin^{-1} b$

⇒ B

11. Trigonometry, EXT1 T1 2024 HSC 4 MC

$$\text{Domain: } -1 \leq 2x \leq 1 \Rightarrow -\frac{1}{2} \leq x \leq \frac{1}{2}$$

$$\text{Range: } 2\left(\cos^{-1}(2x) + \sin^{-1}(2x)\right) = 2 \times \frac{\pi}{2} = \pi$$

⇒ A

12. Trigonometry, EXT1 T1 EQ-Bank 3 MC

$$y = \cos^{-1}(x - 3)$$

Dilate horizontally with scale factor $\frac{1}{2}$

Swap: $x \rightarrow 2x$

$$y_1 = \cos^{-1}(2x - 3)$$

Translate to the left by 1

Swap: $x \rightarrow x + 1$

$$y_2 = \cos^{-1}(2(x + 1) - 3)$$

$$= \cos^{-1}(2x + 2 - 3)$$

$$= \cos^{-1}(2x - 1)$$

⇒ C

13. Trigonometry, EXT1 T1 EQ-Bank 4 MC

$$y = \sin^{-1}(x - 2)$$

Dilate by factor 3 from the y -axis

$$\text{Swap: } x \rightarrow \frac{x}{3}$$

$$\begin{aligned} y_1 &= \sin^{-1}\left(\frac{x}{3} - 2\right) \\ &= \sin^{-1}\left(\frac{x - 6}{3}\right) \end{aligned}$$

Translate to the right by 2

$$\text{Swap: } x \rightarrow x - 2$$

$$\begin{aligned} y_2 &= \sin^{-1}\left(\frac{(x - 2) - 6}{3}\right) \\ &= \sin^{-1}\left(\frac{x - 8}{3}\right) \end{aligned}$$

$$\Rightarrow A$$

14. Trigonometry, EXT1 T1 EQ-Bank 5 MC

Graph shape (general) $\rightarrow \cos^{-1}(x)$

Graph translated 1 unit to the right $\rightarrow \cos^{-1}(x - 1)$

$$\text{Range: } 0 \leq \cos^{-1}(x - 1) \leq \pi \rightarrow 0 \leq 2\cos^{-1}(x - 1) \leq 2\pi$$

$$\Rightarrow C$$

15. Trigonometry, EXT1 T1 EQ-Bank 2

$$\text{Consider domain: } -1 \leq 3x \leq 1 \Rightarrow -\frac{1}{3} \leq x \leq \frac{1}{3}$$

$$\text{Domain: } \left[-\frac{1}{3}, \frac{1}{3}\right]$$

$$\text{Range: } [0, \pi]$$

16. Trigonometry, EXT1 T1 2011 HSC 1e

$$\cos \frac{\pi}{3} = \frac{1}{2}$$

Since $\cos$ is negative in 2nd quadrant,

$$\cos^{-1}\left(-\frac{1}{2}\right) = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$$

17. Trigonometry, EXT1 T1 2005 HSC 1c

$$y = \cos^{-1} \frac{x}{4}$$

Domain of $y = \cos^{-1}x$ is

$$-1 \leq x \leq 1$$

$\therefore$ Domain of $y = \cos^{-1} \frac{x}{4}$ is

$$-1 \leq \frac{x}{4} \leq 1$$

$$-4 \leq x \leq 4$$

Range $y = \cos^{-1}x$ is

$$0 \leq y \leq \pi$$

$\therefore$ Range $y = \cos^{-1} \frac{x}{4}$ is

$$0 \leq y \leq \pi$$

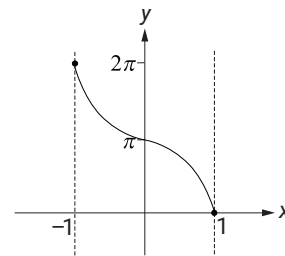
18. Trigonometry, EXT1 T1 2017 HSC 11d

$$y = 2\cos^{-1}x$$

$$\Rightarrow \text{Domain } -1 \leq x \leq 1$$

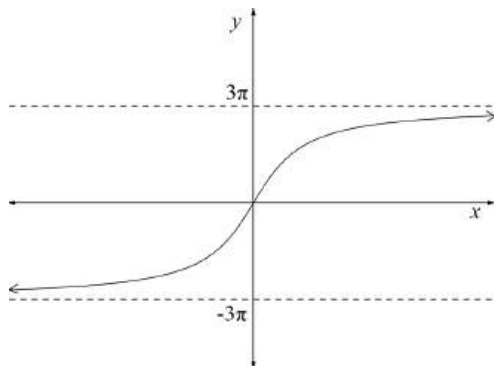
$$\text{Range } y = 2\cos^{-1}x \text{ is } 0 \leq y \leq \pi$$

$$\Rightarrow \text{Range } y = 2\cos^{-1}x \text{ is } 0 \leq y \leq 2\pi$$

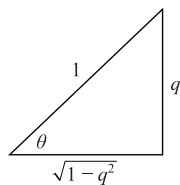


19. Trigonometry, EXT1 T1 2014 HSC 11c

$$y = 6 \tan^{-1} x$$



20. Trigonometry, EXT1 T1 EQ-Bank 1



$$\sin \theta = q$$

$$\theta = \sin^{-1} q$$

$$\begin{aligned} \therefore \cos(\sin^{-1} q) &= \cos \theta \\ &= \sqrt{1 - q^2} \end{aligned}$$

21. Trigonometry, EXT1 T1 SM-Bank 2

$$\text{Base angle: } \cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}$$

$\cos \Rightarrow$ -ve in 2nd quadrant

$$\begin{aligned} \cos^{-1} \left(-\frac{\sqrt{3}}{2} \right) &= \pi - \frac{\pi}{6} \\ &= \frac{5\pi}{6} \end{aligned}$$

22. Calculus, EXT1 C2 2006 HSC 2a

i. $f(x) = \sin^{-1}(x + 5)$

Domain

$$-1 \leq x + 5 \leq 1$$

$$-6 \leq x \leq -4$$

Range

$$-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$$

ii. $y = \sin^{-1}(x + 5)$

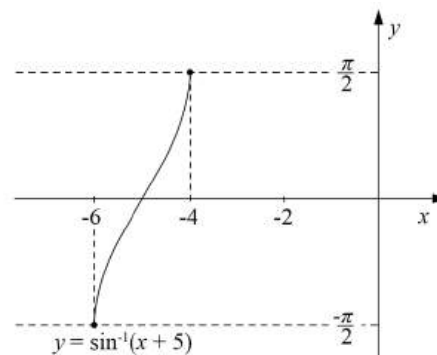
$$\frac{dy}{dx} = \frac{1}{\sqrt{1 - (x + 5)^2}}$$

When $x = -5$

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{\sqrt{1 - (-5 + 5)^2}} \\ &= \frac{1}{\sqrt{1 - 0}} \\ &= 1 \end{aligned}$$

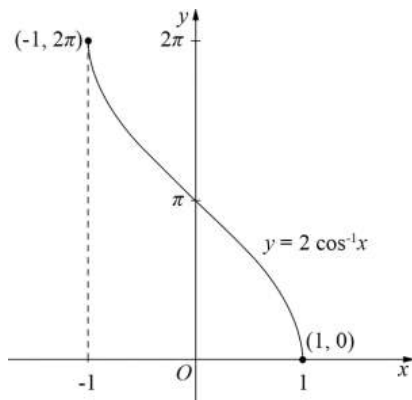
$\therefore$ Gradient of $y = f(x)$ at $x = -5$ is 1.

iii.



23. Trigonometry, EXT1 T1 2007 HSC 2b

- i. $y = 2 \cos^{-1} x$
 Domain: $-1 \leq x \leq 1$
 Range: $0 \leq \frac{y}{2} \leq \pi$
 $0 \leq y \leq 2\pi$



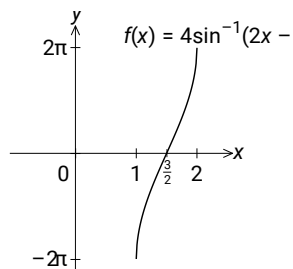
- ii. Range: $0 \leq y \leq 2\pi$

24. Trigonometry, EXT1 T1 EQ-Bank 5

- a. Domain: $-1 \leq 2x - 3 \leq 1$
 $2 \leq 2x \leq 4$
 $1 \leq x \leq 2$

Range: $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$
 $-2\pi \leq 4y \leq 2\pi$

- b.



25. Trigonometry, EXT1 T1 2007 HSC 5c

$$\sin^{-1} x + \frac{1}{2} \cos^{-1} y = \frac{\pi}{3} \quad \dots (1)$$

$$3 \sin^{-1} x - \frac{1}{2} \cos^{-1} y = \frac{2\pi}{3} \quad \dots (2)$$

Add (1) + (2)

$$4 \sin^{-1} x = \pi$$

$$\sin^{-1} x = \frac{\pi}{4}$$

$$\therefore x = \frac{1}{\sqrt{2}}$$

Substitute $x = \frac{1}{\sqrt{2}}$ into (1)

$$\sin^{-1} \frac{1}{\sqrt{2}} + \frac{1}{2} \cos^{-1} y = \frac{\pi}{3}$$

$$\frac{\pi}{4} + \frac{1}{2} \cos^{-1} y = \frac{\pi}{3}$$

$$\frac{1}{2} \cos^{-1} y = \frac{\pi}{12}$$

$$\cos^{-1} y = \frac{\pi}{6}$$

$$\therefore y = \frac{\sqrt{3}}{2}$$

MARKER'S COMMENT: The use of the elimination method here proved much more successful than substitution.

26. Trigonometry, EXT1 T1 2010 HSC 1b

$$f(x) = \cos^{-1} \left(\frac{x}{2} \right)$$

$$-1 \leq \frac{x}{2} \leq 1$$

$$-2 \leq x \leq 2$$

$\therefore$ Domain of $f(x)$ is $-2 \leq x \leq 2$

27. Trigonometry, EXT1 T1 2011 HSC 2d

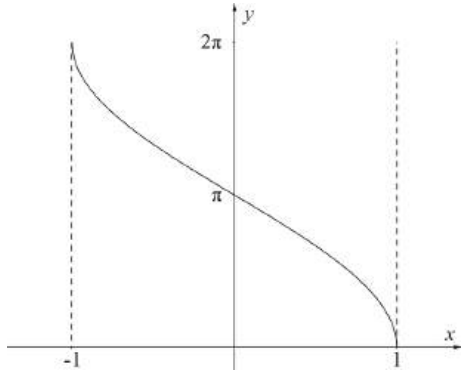
$$y = 2\cos^{-1}x$$

$$\frac{y}{2} = \cos^{-1}x$$

$$\text{Domain } -1 \leq x \leq 1$$

$$\text{Since } 0 \leq \frac{y}{2} \leq \pi$$

$$\text{Range } 0 \leq y \leq 2\pi$$

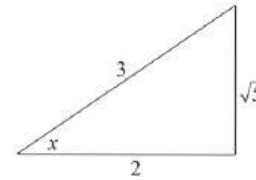


28. Trigonometry, EXT1 T1 2012 HSC 13a

$$\sin\left(2\cos^{-1}\left(\frac{2}{3}\right)\right)$$

$$\text{Let } x = \cos^{-1}\left(\frac{2}{3}\right)$$

$$\therefore \cos x = \frac{2}{3}$$



$$\sin x = \frac{\sqrt{5}}{3}$$

$$\sin\left(2\cos^{-1}\left(\frac{2}{3}\right)\right) = \sin 2x$$

$$= 2\sin x \cos x$$

$$= 2 \times \frac{\sqrt{5}}{3} \times \frac{2}{3}$$

$$= \frac{4}{9}\sqrt{5}$$

TIP: This question is made less complicated by reminding yourself that $\cos^{-1}\left(\frac{2}{3}\right)$ is simply an angle.

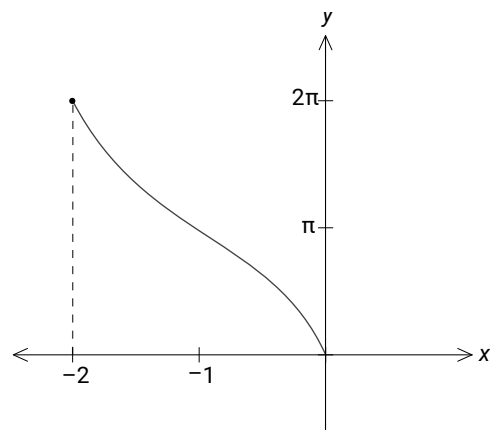
29. Trigonometry, EXT1 T1 EQ-Bank 4

Domain:

$$-1 \leq x + 1 \leq 1 \Rightarrow -2 \leq x \leq 0$$

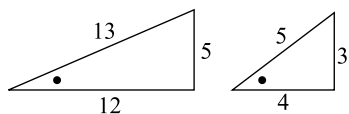
Range:

$$0 \leq \cos^{-1}(x) \leq \pi \Rightarrow 0 \leq 2\cos^{-1}(x) \leq 2\pi$$



30. Trigonometry, EXT1 T1 SM-Bank 1

$$\begin{aligned} \sin \theta &= \sin \left(\cos^{-1} \left(\frac{12}{13} \right) + \tan^{-1} \left(\frac{3}{4} \right) \right) \\ &= \sin \left(\cos^{-1} \left(\frac{12}{13} \right) \right) \cos \left(\tan^{-1} \left(\frac{3}{4} \right) \right) + \sin \left(\tan^{-1} \left(\frac{3}{4} \right) \right) \cos \left(\cos^{-1} \left(\frac{12}{13} \right) \right) \end{aligned}$$



$$\begin{aligned} \therefore \sin \theta &= \frac{5}{13} \times \frac{4}{5} + \frac{3}{5} \times \frac{12}{13} \\ &= \frac{20}{65} + \frac{36}{65} \\ &= \frac{56}{65} \end{aligned}$$

COMMENT: Syllabus states that students must understand and use the notation `arccos`, `arcsin` etc..

31. Trigonometry, EXT1 T1 SM-Bank 3

Base angle: $\sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$

$\sin \Rightarrow$ -ve in 4th quadrant

$$\sin^{-1} \left(\frac{\sqrt{3}}{2} \right) = \frac{\pi}{3}$$

$$\sin^{-1} \left(-\frac{\sqrt{3}}{2} \right) = -\frac{\pi}{3}$$

Base angle: $\cos \frac{\pi}{3} = \frac{1}{2}$

$\cos \Rightarrow$ -ve in 2nd quadrant

$$\cos^{-1} \left(-\frac{1}{2} \right) = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$$

$$\begin{aligned} \therefore \sin^{-1} \left(\frac{\sqrt{3}}{2} \right) - \sin^{-1} \left(-\frac{\sqrt{3}}{2} \right) - \cos^{-1} \left(-\frac{1}{2} \right) \\ &= \frac{\pi}{3} - \left(-\frac{\pi}{3} \right) - \frac{2\pi}{3} \\ &= 0 \end{aligned}$$

32. Trigonometry, EXT1 T1 SM-Bank 4

Domain:

$$-1 \leq (2x - 1) \leq 1$$

$$0 \leq 2x \leq 2$$

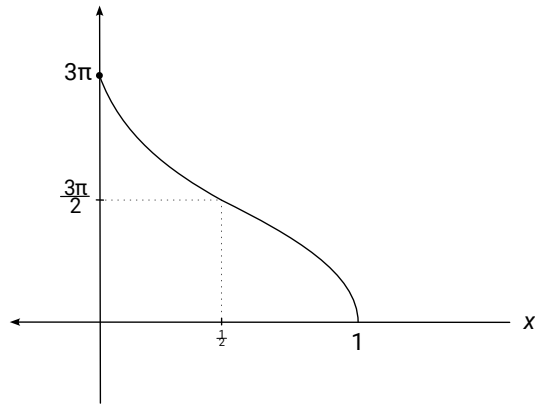
$$0 \leq x \leq 1$$

Range of $\cos^{-1}(x) = [0, \pi]$

$$\Rightarrow \text{Range of } 3\cos^{-1}(x) = [0, 3\pi]$$

At $x = 0$, $y = 3\cos^{-1}(-1) = 3\pi$

Sketch $y = 3\cos^{-1}(2x - 1)$:



33. Trigonometry, EXT1 T1 2022 HSC 13c

The domain of $f(x) = \mathbb{R}$

The range of $g(x) = \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

♦♦ Mean mark 24%.

Since the domain $f(x) \neq$ range of $g(x)$,

$$f^{-1}(x) \neq g(x)$$